Ray tracing – Curved quadratic elements

1. Tetrahedron and ray

Surface position on the tetrahedron's face (weighted sum of shape functions):

$$\boldsymbol{P}(g,h) = \sum_{i=0}^{5} N_i(g,h) \cdot \boldsymbol{f_{nodes_i}} \tag{1}$$

Ray:

$$\boldsymbol{R}(t) = \boldsymbol{o} + t\boldsymbol{d} \tag{2}$$

where $\boldsymbol{d}$ is the ray's direction, and $\boldsymbol{o}$ is the ray's origin.

2. Intersection by minimising vector difference

2.1. Intersection problem

Face-ray intersection (3 unknowns) presented as the vector difference [1, 2]:

$$\boldsymbol{o} + t\boldsymbol{d} - \boldsymbol{P}(g,h) = 0 \tag{3}$$

Nonlinear system of equations:

$$\boldsymbol{F}(t,g,h) = \boldsymbol{o} + t\boldsymbol{d} - \sum_{i-0}^{5} N_i(g,h) \cdot \boldsymbol{f_{nodes_i}} = \boldsymbol{0} \tag{4}$$

The system of equations contains 3 unknowns and 3 equations, so the mapping is:

$$\Re^3 \rightarrow \Re^3 \tag{5}$$

They represent the distance vector between a point on the ray and a point on the surface.

2.2. Möller-Trumbore algorithm for linear triangle intersection (initial guess)

$$\boldsymbol{E_1} = V_1 - V_0,\ \boldsymbol{E_2} = V_2 - V_0 \tag{6}$$

$$\boldsymbol{p} = \boldsymbol{d} \times \boldsymbol{E_2} \tag{7}$$

$$det = \boldsymbol{E_1} \cdot \boldsymbol{p} \tag{8}$$

$$\boldsymbol{t_{vec}} = \boldsymbol{o} - V_0 \tag{9}$$

Where V_0, V_1, V_2 are triangle vertices.

$$u = \frac{\boldsymbol{t_{vec}} \cdot \boldsymbol{p}}{det} \tag{10}$$

$$\boldsymbol{q} = \boldsymbol{t_{vec}} \times \boldsymbol{E_1} \tag{11}$$

$$v = \frac{\boldsymbol{d} \cdot \boldsymbol{q}}{det} \tag{12}$$

$$t = \frac{\boldsymbol{E_2} \cdot \boldsymbol{q}}{det} \tag{13}$$

2.3. Newton-Raphson method

Since there are 3 variables and 3 equations, Newton-Raphson method could be used to find the solution. Only Jacobians of F are required.

Jacobian of F:

$$\boldsymbol{J_F} = \left[\frac{\partial \boldsymbol{F}}{\partial f}, \frac{\partial \boldsymbol{F}}{\partial g}, \frac{\partial \boldsymbol{F}}{\partial h}\right] \tag{14}$$

$$\boldsymbol{J_F} = \left[\boldsymbol{d}, -\frac{\partial \boldsymbol{P}}{\partial g}, -\frac{\partial \boldsymbol{P}}{\partial h}\right] \tag{15}$$

$$\frac{\partial \boldsymbol{P}}{\partial g} = \sum_{i=0}^{5} \frac{\partial N_i}{\partial g} \cdot \boldsymbol{f_{nodes_i}} \tag{16}$$

$$\frac{\partial \boldsymbol{P}}{\partial h} = \sum_{i=0}^{5} \frac{\partial N_i}{\partial h} \cdot \boldsymbol{f_{nodes_i}} \tag{17}$$

The update by solving a linear system:

$$\boldsymbol{\Delta} = [\Delta t, \Delta g, \Delta h]^T \tag{18}$$

$$\boldsymbol{J_F} \cdot \boldsymbol{\Delta} = -\boldsymbol{F}(t, g, h) \tag{19}$$

$$\begin{bmatrix} d_x & -\frac{\partial P_x}{\partial g} & -\frac{\partial P_x}{\partial h} \\ d_y & -\frac{\partial P_y}{\partial g} & -\frac{\partial P_y}{\partial h} \\ d_z & -\frac{\partial P_z}{\partial g} & -\frac{\partial P_z}{\partial h} \end{bmatrix} \begin{bmatrix} \Delta t \\ \Delta g \\ \Delta h \end{bmatrix} = -(\boldsymbol{o} + t\boldsymbol{d} - \boldsymbol{P}(g, h)) \tag{20}$$

3. Intersection by minimising scalar squared distance

3.1. Intersection problem

Face-ray intersection could be presented as the scalar squared difference [3]:

$$F(g, h) = |\boldsymbol{V}|^2 - (\boldsymbol{V} \cdot \boldsymbol{d})^2 \tag{21}$$

where

$$\boldsymbol{V} = \boldsymbol{P}(g, h) - \boldsymbol{o} \tag{22}$$

and

$$t = \boldsymbol{V} \cdot \boldsymbol{d} = (\boldsymbol{P}(g, h) - \boldsymbol{o}) \cdot \boldsymbol{d} \tag{23}$$

The equation contains 2 unknowns, so the mapping is:

$$\Re^2 \rightarrow \Re^1 \tag{24}$$

They represent the squared perpendicular distance between the ray and surface (Figure 1).

If a local minimum of F is zero, then this corresponds to a point where the ray intersects the surface. Those points where F is a minimum, but where $F > 0$, indicate that the ray misses the surface by a finite distance. This is actually desirable in that the algorithm will still converge near "silhouette edges" of surface, but will give a non-zero minimum.

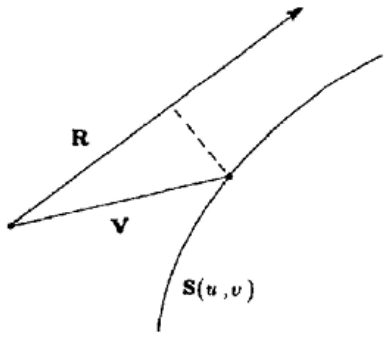

Figure 1: Distance from a point on the surface to a ray [3], S is P in this document.

3.2. Quasi-Newton method

It is not possible to use Newton-Raphson method in this case, as there are 3 variables and 1 equation. Additionally, this equation cannot be represented as a sum of squared residuals to be solved by Gauss-Newton approach. Consequently, it is necessary to know Jacobians as well as Hessians of F.

Jacobian of F:

$$\boldsymbol{J_F} = \left[\frac{\partial F}{\partial g}, \frac{\partial F}{\partial h}\right]^T \tag{25}$$

Let:

$$\boldsymbol{P}_g = \frac{\partial \boldsymbol{P}}{\partial g} \tag{26}$$

$$\boldsymbol{P}_h = \frac{\partial \boldsymbol{P}}{\partial h} \tag{27}$$

$$\boldsymbol{P}_{gg} = \frac{\partial^2 \boldsymbol{P}}{\partial g^2} \tag{28}$$

$$\boldsymbol{P}_{hh} = \frac{\partial^2 \boldsymbol{P}}{\partial h^2} \tag{29}$$

$$\boldsymbol{P}_{gh} = \boldsymbol{P}_{hg} = \frac{\partial^2 \boldsymbol{P}}{\partial gh} = \frac{\partial^2 \boldsymbol{P}}{\partial hg} \tag{30}$$

The partials of F are:

$$\frac{\partial F}{\partial g}(g,h) = 2\left(\frac{\partial \boldsymbol{P}}{\partial g}\cdot\boldsymbol{V} - t\frac{\partial \boldsymbol{P}}{\partial g}\cdot\boldsymbol{d}\right) = \tag{31}$$

$$= 2\frac{\partial \boldsymbol{P}}{\partial g}\left(\boldsymbol{V} - t\boldsymbol{d}\right) = \tag{32}$$

$$= 2\frac{\partial \boldsymbol{P}}{\partial g}\left(\boldsymbol{P} - (\boldsymbol{o} + t\boldsymbol{d})\right) \tag{33}$$

$$\frac{\partial F}{\partial h}(g,h) = 2\frac{\partial \boldsymbol{P}}{\partial h}\left(\boldsymbol{P} - (\boldsymbol{o} + t\boldsymbol{d})\right) \tag{34}$$

$$\frac{\partial F}{\partial g}(g,h) = 2\boldsymbol{P}_g\left(\boldsymbol{V} - t\boldsymbol{d}\right) \tag{35}$$

$$\frac{\partial F}{\partial h}(g,h) = 2\boldsymbol{P}_h\left(\boldsymbol{V} - t\boldsymbol{d}\right) \tag{36}$$

Hessian of F:

$$\boldsymbol{H_F} = \begin{bmatrix} \frac{\partial^2 F}{\partial g^2} & \frac{\partial^2 F}{\partial g h} \\ \frac{\partial^2 F}{\partial h g} & \frac{\partial^2 F}{\partial h^2} \end{bmatrix} \tag{37}$$

The analytical formulae could be derived to calculate precise $\boldsymbol{H_F}$; however, the calculation process is quite time-consuming.

$$\frac{\partial F}{\partial g^2} = 2\left[|\boldsymbol{P}_g|^2 - (\boldsymbol{P}_g\cdot\boldsymbol{d})^2 + \boldsymbol{V}\cdot\boldsymbol{P}_{gg} - (\boldsymbol{V}\cdot\boldsymbol{d})(\boldsymbol{P}_{gg}\cdot\boldsymbol{d})\right] \tag{38}$$

$$\frac{\partial F}{\partial h^2} = 2\left[|\boldsymbol{P}_h|^2 - (\boldsymbol{P}_h\cdot\boldsymbol{d})^2 + \boldsymbol{V}\cdot\boldsymbol{P}_{hh} - (\boldsymbol{V}\cdot\boldsymbol{d})(\boldsymbol{P}_{hh}\cdot\boldsymbol{d})\right] \tag{39}$$

$$\frac{\partial F}{\partial gh} = 2\left[\boldsymbol{P}_g\cdot\boldsymbol{P}_h - (\boldsymbol{P}_g\cdot\boldsymbol{d})(\boldsymbol{P}_h\cdot\boldsymbol{d}) + \boldsymbol{V}\cdot\boldsymbol{P}_{gh} - (\boldsymbol{V}\cdot\boldsymbol{d})(\boldsymbol{P}_{gh}\cdot\boldsymbol{d})\right] \tag{40}$$

The update by solving a linear system:

$$\boldsymbol{x} = [g,h]^T \tag{41}$$

$$\boldsymbol{\Delta} = [\Delta g, \Delta h]^T \tag{42}$$

$$\boldsymbol{\Delta} = -s\boldsymbol{H_F}^{-1}\boldsymbol{J_F} \tag{43}$$

$$\boldsymbol{H_F}\cdot\boldsymbol{\Delta} = -s\boldsymbol{J_F} \tag{44}$$

$$\begin{bmatrix} \frac{\partial^2 F}{\partial g^2} & \frac{\partial^2 F}{\partial gh} \\ \frac{\partial^2 F}{\partial hg} & \frac{\partial^2 F}{\partial h^2} \end{bmatrix} \begin{bmatrix} \Delta g \\ \Delta h \end{bmatrix} = -s \begin{bmatrix} \frac{\partial F}{\partial g}, \frac{\partial F}{\partial h} \end{bmatrix}^T \tag{45}$$

where s is a scalar satisfying the following relationship:

$$s = \min_s \left(F \left(\boldsymbol{x} - s \boldsymbol{H}_{\boldsymbol{F}}^{-1} \boldsymbol{J}_{\boldsymbol{F}} \right) \right) \tag{46}$$

The scalar s can be found by employing backtracking line search with Armijo condition in the following direction [4]:

$$\boldsymbol{p} = -\boldsymbol{H}_{\boldsymbol{F}}^{-1} \boldsymbol{J}_{\boldsymbol{F}} \tag{47}$$

The directional derivative is:

$$D_p = \boldsymbol{J}_{\boldsymbol{F}} \cdot \boldsymbol{p} \tag{48}$$

Iterate $\boldsymbol{x}$ as:

$$\boldsymbol{x} = \boldsymbol{x}_{\boldsymbol{init}} + s\boldsymbol{p} \tag{49}$$

$$s = s\beta \tag{50}$$

Iterate until Armijo condition is fulfilled:

$$F(\boldsymbol{x}) \leq F(\boldsymbol{x}_{\boldsymbol{init}}) + csD_p \tag{51}$$

4. Surface normal

$$\boldsymbol{n} = \frac{\partial \boldsymbol{P}}{\partial g} \times \frac{\partial \boldsymbol{P}}{\partial h} \tag{52}$$

References

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[2] D. L. Toth, On ray tracing parametric surfaces, SIGGRAPH Comput. Graph. 19 (1985) 171–179. URL: https://doi.org/10.1145/325165.325233. doi:10.1145/325165.325233.

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